

# Four findings from a solo open-web LLM red-team, the negatives included.

Every result here is measured and reproducible. An LLM-as-judge calibrated against human labels and then generalized into a four-breach-type discipline; scheduling shown to be a capability lever, not just an optimization; a publication-grade null result that redirected engineering; and the measure-before-build habit behind all of it.

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## 01 JUDGE

### Calibrating an LLM-as-judge against human labels, then recalibrating when a benchmark exposed it.

Every breach verdict is an LLM judgment, so the judge is the load-bearing weakness. It was validated four ways, three against independent human-annotated benchmarks: blind stratified in-distribution hand-labels, WildGuardTest (Allen AI annotators), StrongREJECT, and JailbreakBench's *judge\_comparison* (300 human-labeled rows against four field classifiers). JailbreakBench exposed over-flagging: the v1 judge agreed with the human majority only **70.3%, last of five** (behind HarmBench, LlamaGuard-2, GPT-4, Llama-3), at recall 98% and precision 55%.

A 20-row false-positive audit traced it to **five recurring failure modes**. The root cause was a *rubric* problem: it rewarded engagement with the attack frame (persona acceptance, acknowledgment, format mimicry) over transfer of harmful content. A **content-transfer-gate rubric (v3)** moved the same 300 rows to **89.3% agreement, 79.5% precision, 95.5% recall** (a 19-point agreement gain, 24.5-point precision gain, 2.5-point recall cost), lifting it from last to 3rd of five and tied with the frontier classifiers, for about \$8.4 via a tiered evaluation.

Then the honest part. Re-judging the stored breach matrix under v3 cut breach cells from **2,429 to 1,371, a 43.6% reduction**, correcting prior over-reporting. All three external axes were re-measured under v3 (WildGuard harm 88.5%; StrongREJECT 26% more conservative).

**70.3 → 89.3%**

JBB HUMAN AGREEMENT

**55 → 79.5%**

PRECISION

**2,429 → 1,371**

BREACH CELLS  
RE-JUDGED

**2.56%**

IN-DIST FALSE-POSITIVE



Judge agreement vs the field (JailbreakBench, % vs the human majority). ROGUE v1 last; v3 tied with the frontier classifiers.

WHY THIS IS NOTABLE

A named false-positive taxonomy for a safety judge, plus a measured finding that two respected benchmarks (WildGuardTest harm labels and StrongREJECT) themselves *over-count* relative to a strict content-transfer standard.

**The gate is not harm-specific, it is a calibration discipline.** An established practice taken rigorously, not a new method. One consummation-gate template, re-instantiated per breach type, now calibrates **four structurally different breach classes**: a *harm* breach (capability transfer, re-confirmed at **91.0% agreement, 273 of 300**, top-of-field); a *content* breach (information-disclosure, did the protected datum appear), whose v2 rubric defines a materially-revealing threshold and reaches **97.35% agreement**, 100% recall, 0% FP-mode (up from 94.74%); an *action* breach (unauthorized-action, did the agent execute), whose v3 tool-trace rubric reaches **98.89% agreement**, 100% recall, with FP-mode driven 9.38% to 6.25% to 3.12%; and a new *fabrication* breach (fabricated-sensitive-value, a trust breach distinct from disclosure) at **100% agreement**, 0% FP-mode on the first pass. The harness **self-diagnoses**: on the harder types it returned *refine*, a targeted rubric fix was applied, and re-measurement shipped it, the same diagnose then fix then re-measure loop that produced v3, run again. The tool-trace turned a stated limitation into a measured resolution: the action type's earlier weakness was the text-only proxy, not the gate, so a tool-call trace makes "executed" a fact and dissolves the simulate-or-claim confusion. This pattern generalizes: provenance-dependent breach types need an evidence trace, not a better rubric, and the independence check names the missing evidence, shown twice, a tool-call trace that lifted second-labeler kappa from 0.746 to 0.917 for unauthorized-action and a retrieval trace that lifted kappa from 0.723 to 0.909 for fabricated-value. The contribution is **a repeatable discipline for calibrating breach judges across breach classes**, not a single judge.

|                                        |                              |                                            |                                |
|----------------------------------------|------------------------------|--------------------------------------------|--------------------------------|
| 91.0%<br>HARM (CAPABILITY<br>TRANSFER) | 97.35%<br>INFO-DISCLOSURE V2 | 98.89%<br>UNAUTH-ACTION V3<br>(TOOL-TRACE) | 100%<br>FABRICATED-VALUE (NEW) |
|----------------------------------------|------------------------------|--------------------------------------------|--------------------------------|

#### WHY THIS IS NOTABLE

Four breach classes, one gate template, every variant shipped. The methodology exposes type-dependent difficulty: action consummation (did the agent execute) was the hardest, and the tool-call trace resolved it by making execution a recorded fact rather than a text-only proxy. Caveats stated plainly: single-operator kappa (the tool-trace lifted unauthorized-action from 0.746 to 0.917; the fabricated-value retrieval-trace lifted its human kappa from 0.723 to 0.909); corpora are synthetic. The building blocks are established (trace-grounded agent eval, kappa-gated calibration, provenance attribution, cross-type judge generalization like CompliBench); the contribution is the rigor and the measured cross-type result, not a new mechanism. These are descriptive measurements, not validated generalizations.

## 02

### SCHEDULING

## Scheduling as a capability lever, not just an optimization.

A within-tier greedy reorder was replaced with a **target-conditioned cross-tier scheduler**: a static, explainable blend (0.5 global, 0.3 vendor, 0.2 family breach-rate; Laplace-smoothed; deliberately no ML and no bandit, so it stays reproducible). A single-variable controlled experiment (same ladder, attacks, corpus, judge, and target on Claude Haiku across AdvBench and JailbreakBench, with only the order changed) beat the production baseline on every axis: **median winner-rank 22 → 11 to 13.5**, **attack-success-rate 50% → 60%**, and **cost-per-success \$1.25 → \$0.74 (41% cheaper)**.

**22 → 11**

MEDIAN WINNER-RANK

**50 → 60%**

ATTACK-SUCCESS-RATE

**41% cheaper**

COST PER SUCCESS

#### WHY THIS IS NOTABLE

The mechanism is the interesting part: a lower rank *caused* a higher success rate. The old order exhausted the per-scan budget cap before reaching the winning technique, so reordering improved coverage, cost, and latency at once with zero new attacks. The reproducibility invariant is “reorder, never exclude”: same ladder, different order, full reachability preserved.

## 03

### NULL RESULT

## A publication-grade null result: grammar-component predictive power.

Before building a grammar/AST attack-composition engine, a **\$0 observational study over 1,540 (primitive × target) cells** tested whether grammar-structure nodes predict breach *beyond* attack-family membership, with full confound controls: Benjamini-Hochberg FDR across hundreds of node and pair tests, Mantel-Haenszel stratification by target model, within-family lift, and Cramér’s-V collinearity flagging. The verdict was **weak to none**: the family label carries the predictive weight, cross-family structural nodes show roughly 1.0 to 1.1× non-significant lift, and the striking pre-FDR pairwise synergies (odds ratios up to 16.8) survived **none** of the four controls.

1,540

CELLS, \$0 STUDY

~1.0 to 1.1×

CROSS-FAMILY LIFT (N.S.)

0 of 4

SYNERGIES SURVIVED CONTROLS

#### WHY THIS IS NOTABLE

A cheap, rigorous falsification that redirected engineering away from a months-long build: a successful negative result.

## 04

DISCIPLINE

### Measure-before-build discipline.

\$0 measurements from existing telemetry were used repeatedly to *invert* “build it” decisions, each parked with an explicit trigger to revisit:

- **Per-model ladder routing.** The spread was a model main effect, not a family×model interaction, so it was not worth the rewrite.
- **LLM renderer-synthesis.** The synthesis-grade backlog stayed flat at 7 across two widening harvests, so it was parked.
- **HF jailbreak-dataset bulk-import.** It measured 0 new attack families, so it was declined.

## 05

REMEDiation

### Measured remediation: prove a fix closes the breach without over-blocking, or refuse to ship it.

Finding a breach is half the job. ROGUE also **generates a candidate fix**, then *measures*, by re-scanning a mutated test config with the same calibrated judge, whether it closes the breach **without over-blocking** legitimate traffic, and **refuses** any it cannot prove (it generates and verifies; the client deploys; it never sits in the request path). Across live runs it **refused every offline patch**, each for a distinct measured reason: a medical/financial-directive patch **did not reduce the breach (20.8% → ~25%)**, and a system-prompt-extraction patch **over-blocked legitimate traffic**, the calibrated over-block judge flagged ~20% where a marker heuristic had scored 0%, so the loop refused it for an architecture change. The “without over-blocking” check is itself calibrated and earned its keep: an over-block judge scored against a 50-case independent set reaches **98% agreement, 100% precision, 0% over-flag** (vs an 88% marker heuristic) and caught the over-block the heuristic missed.

0% → ~20%

RD04 OVER-BLOCK:  
HEURISTIC → JUDGE

20.8 → ~25%

RA06 PATCH: NO  
REDUCTION

98% / 0%

OVER-BLOCK JUDGE:  
AGREE / OVER-FLAG

88 → 98%

OVER-BLOCK DETECTOR  
CALIBRATED

#### WHY THIS IS NOTABLE

The contribution is not a new mitigation but the discipline: a fix is accepted only when a re-scan proves it closes the breach without over-blocking, and refused otherwise, and the calibrated judge is what makes “does not over-block” trustworthy. It flipped a would-be accept (heuristic 0% over-block) into a correct refusal (judge ~20%). A runtime guardrail asserts it blocks; this measures it, and says no when a patch does not hold or over-blocks.

#### LIMITATIONS, STATED PLAINLY

Targets are black-box live-API models whose versions are not pinned. Some cells are small-n (95% bootstrap confidence intervals are persisted precisely because of this). The judge is single-operator-calibrated. These are descriptive measurements of a live system, not validated generalizations.